NAMOGRAM SOCIAL MEDIA PLATFORM

Design Benchmark

Team ID: 6 **Course:** CS-543 Web Services

Team Members

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1. Capability List (The Verbs)

- User Authentication & Profile Management
- Multimedia Post Creation & Publishing
- User Timeline & Content Retrieval
- Interactive Engagement (Like / Unlike)
- Comment Thread Management
- External Media Safety & Moderation Check

2. Service Boundaries & Contracts

Service	Owns	Operations
User Service	User Credentials, Profiles, Auth Tokens	POST /users POST /auth/login GET /users/{id}
Post Service	Posts, Captions, Media Metadata	POST /posts GET /posts/{id} GET /posts
Interaction Service	Likes and Comments	POST /posts/{id}/likes POST /posts/{id}/comments

3. Central Operation Specification: createPost()

Aspect	Details
Inputs	authorId (string), caption (string), mediaUrl (string, required)
Outputs	post_id, authorId, caption, mediaUrl, likesCount, createdAt
Failure Cases	400 Malformed JSON 422 Field Validation Failure 429 Rate Limit Exceeded
Hidden Internal Details	Primary key generation algorithms, DB connection credentials, internal object memory refs.

4. Service Properties Validation Table

Property	Status	Justification / Fix
Reachable	**YES**	Exposed behind well-defined REST HTTP endpoints.
Self-Contained	**YES**	Each service owns its isolated database/store.
Has a Contract	**YES**	Fully defined via OpenAPI 3.0 specification schemas.
Independent	**YES**	Deployable and scalable without cross-service side-effects.
Loosely Coupled	**YES**	Communicates exclusively over HTTP/JSON interfaces.